Data Analytics Internship

At ApexPlanet Software Pvt. Ltd.

TASK 4 :

Data Storytelling & Statistical Validation

PRESENTED BY

Student Name: Onkar Abhiman Londhe

College Name: S.B Patil College Of Engineering Pune

Department: Computer

Email ID: onkarlondhe1139@gmail.com

AGENDA

Task 1: Data Collection & Data Cleaning

Task 2: Exploratory Data Analysis (EDA)

Task 3: Data Visualization & Dashboard

Task 4: Data Storytelling & Statistical Validation

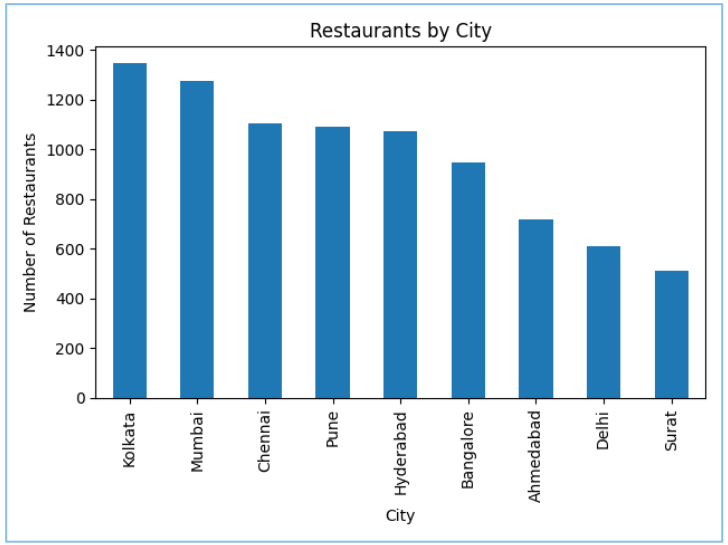
Task 1: Data Collection & Data Cleaning

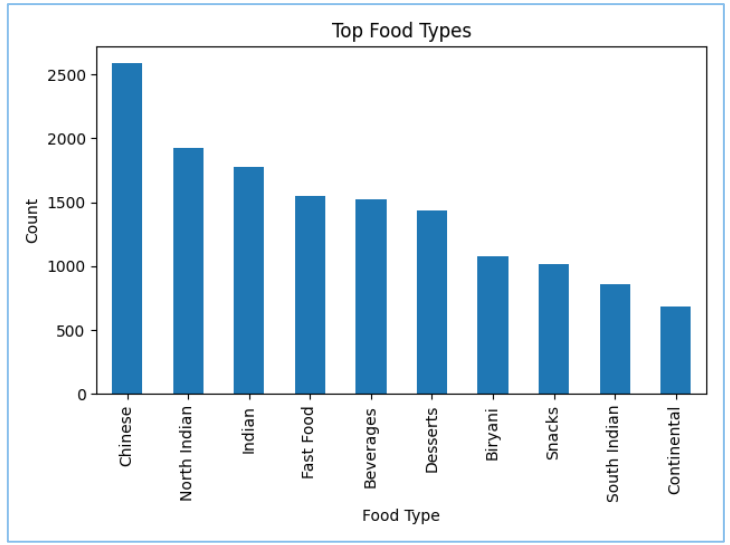
- Collected dataset related to business operations (e.g., product sales dataset).
- Dataset includes multiple attributes such as product details, price, sales metrics and timestamps.
- Goal: Prepare structured data for analysis.

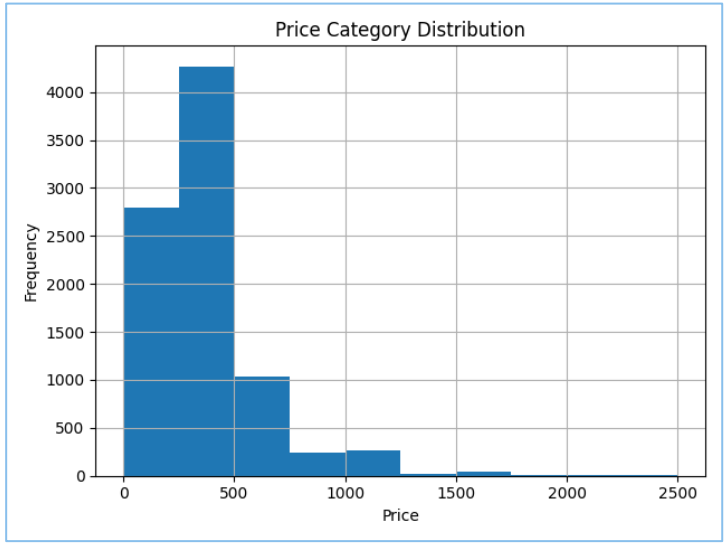
Performed preprocessing to improve data quality:

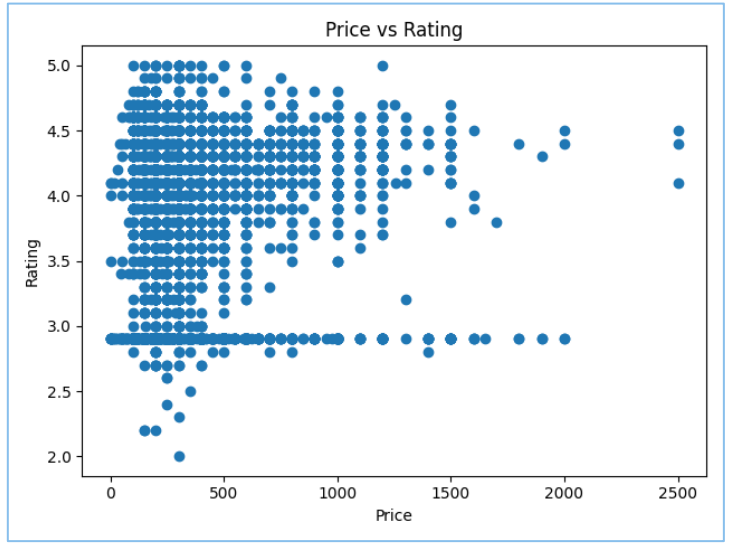
- Removed duplicate records
- Handled missing values
- Standardized text fields
- Removed outliers using statistical techniques (IQR)
- Generated a cleaned dataset for analysis.

Task 1: Data Collection & Data Cleaning









Task 2: Exploratory Data Analysis (EDA)

Analyzed dataset to understand patterns and relationships:

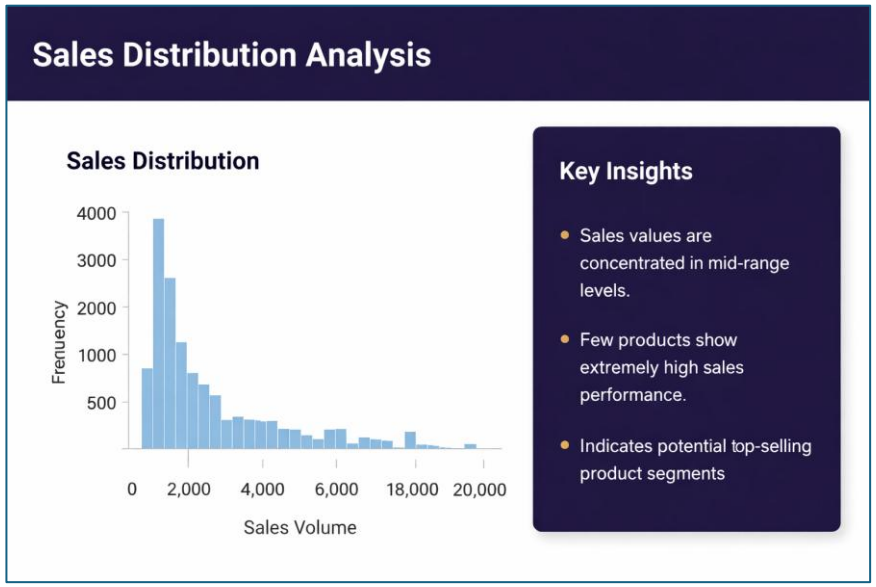
- Price distribution
- Sales volume trends
- Product level performance
- Identification of high-demand items.

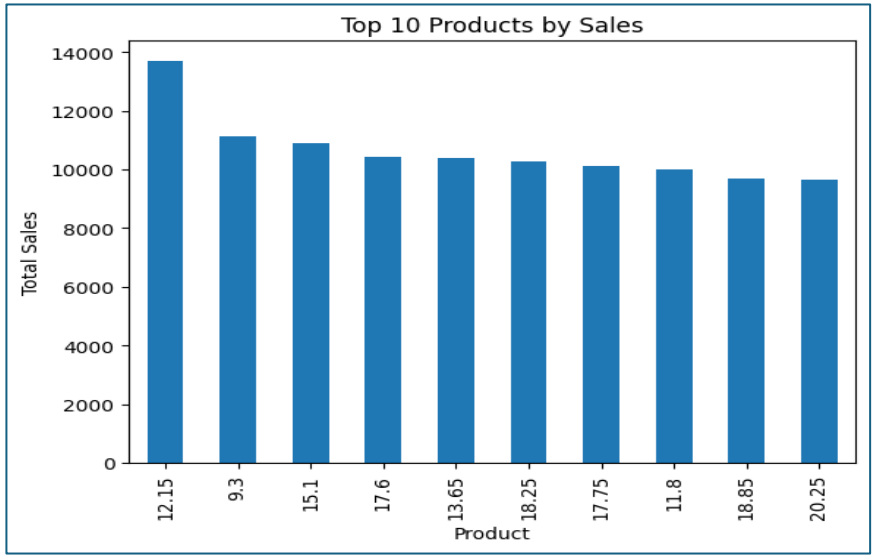
EDA Key Insights

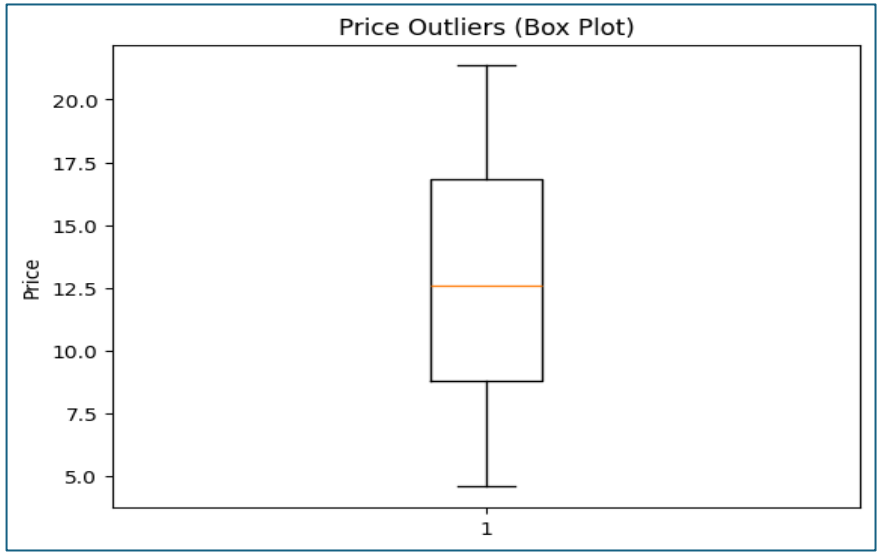
Important insights discovered during analysis:

- Some price ranges generate higher sales
- Certain products dominate overall demand
- Sales distribution shows patterns useful for business decisions.

Task 2: Exploratory Data Analysis (EDA)







Task 3 – Data Visualization

- Created interactive visualizations to communicate insights effectively.
- Tools used: Power BI / visualization tools.

Visuals included:

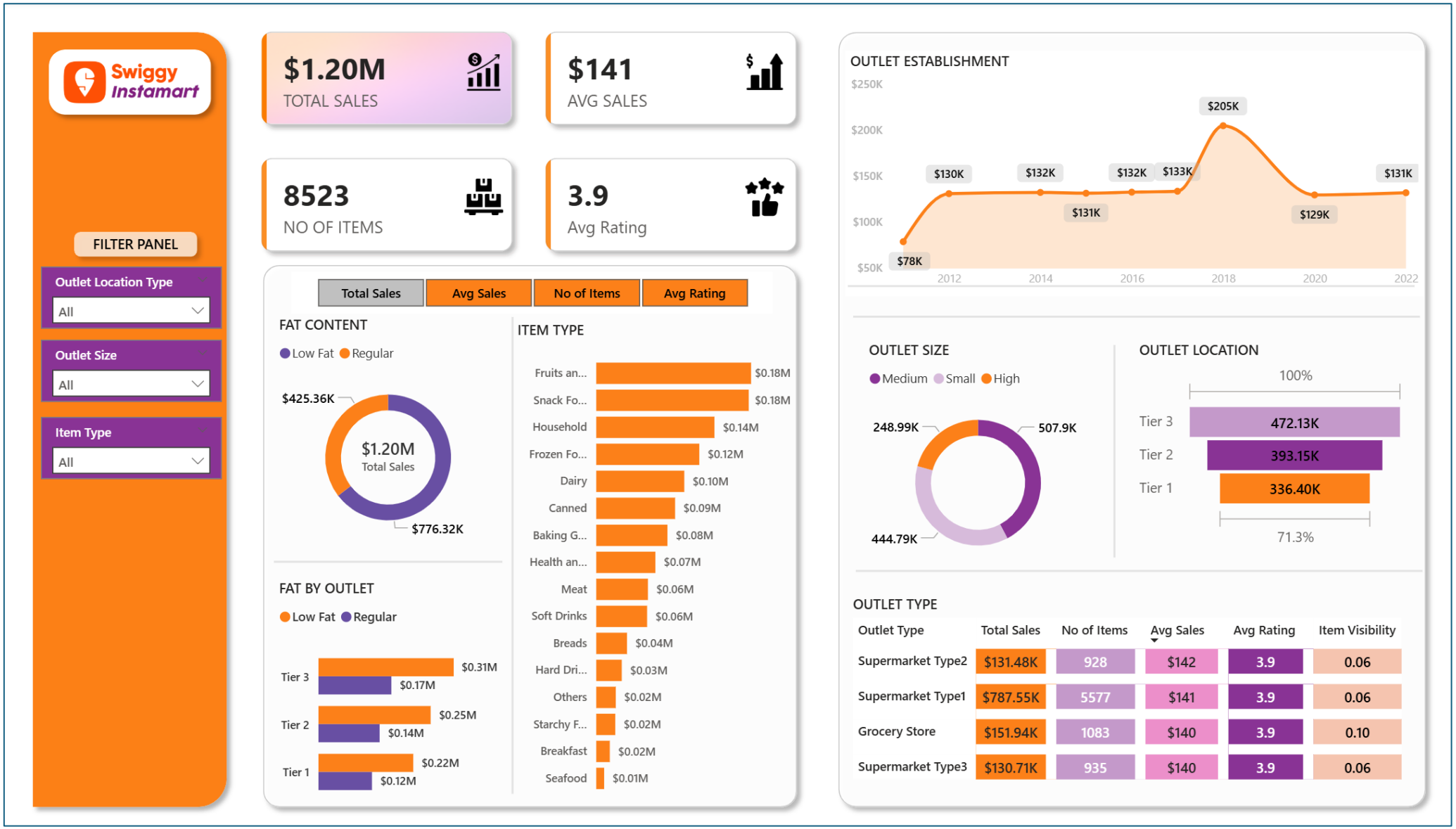
- Sales trends
- Product performance charts
- Price vs demand analysis
- Category comparisons.

Dashboard Benefits

Interactive dashboard enables:

- Real-time insight exploration
- Better business decision making
- Easy identification of trends and patterns
- Clear communication of analytical results.

Task 3 – Swiggy Instamart Sales Dashboard Using Power BI



Outlet Type	Total Sales	No of Items	Avg Sales	Avg Rating	Item Visibility
Supermarket Type2	$131.48K	928	$142	3.9	0.06
Supermarket Type1	$787.55K	5577	$141	3.9	0.06
Grocery Store	$151.94K	1083	$140	3.9	0.10
Supermarket Type3	$130.71K	935	$140	3.9	0.06

Task 4 – Objective

To synthesize all analysis into a compelling business narrative and use basic statistical methods to add rigor and validate key findings.

Step 1 – Craft the Data Story

Structure findings from Tasks 1–3 into a cohesive narrative for business stakeholders.

The story should include:

- Clear objective
- Analytical insights
- Business conclusions
- Recommended actions.

Step 2 – Develop the Final Presentation

Create a professional presentation deck that is:

- Visual
- Concise
- Insight driven
- Focused on business impact.

Step 3 – Hypothesis Testing

Formulate a testable business hypothesis.
Example:
'A change in strategy or design leads to a statistically significant improvement in performance metrics.'

Statistical Testing Methods

Perform appropriate statistical tests such as:

- T-test (A/B testing)
- Chi-Squared test for proportions
- Basic correlation analysis

These tests validate whether patterns observed in the data are statistically significant.

Interpreting Results

Interpret statistical outputs including:

- P-value
- Confidence intervals
- Test statistics

These results determine whether to accept or reject the hypothesis and derive business conclusions.

Hypothesis Testing

Introduction

Statistical validation was used to confirm analytical findings
Significance Level (alpha) = 0.05

Hypothesis Test 1

Test: Price vs Sales Volume
H0: Price does not affect sales
H1: Price affects sales
Method: Correlation Test

Hypothesis Test 2

Test: Discount vs Sales
H0: Discount does not increase sales
H1: Discount increases sales
Method: T-Test

Hypothesis Test 3

Test: Time vs Sales Volume
H0: Sales are equal across time periods
H1: Sales vary across time periods
Method: ANOVA

Deliverables

GitHub:

TASK 4 : https://github.com/onkarlondhe1139/OnkarLondhe-Task4-Hypothesis-Testing

TASK 1 : https://github.com/onkarlondhe1139/OnkarLondhe-Task1-DataWrangling

TASK 2 : https://github.com/onkarlondhe1139/OnkarLondhe-Task-2-EDA-BI

TASK 3 : https://github.com/onkarlondhe1139/OnkarLondhe-Task3-PowerBI-Dashboard

Conclusion

Combining data analysis, visualization, storytelling, and statistical validation enables organizations to make informed, data-driven decisions and clearly communicate insights to stakeholders.

THANK YOU…